



Oracle Database@Azure Automation

Operations Advisory Team

Office of CTO | EMEA Technology Engineering

May 2026

Agenda

1. **Oracle Database at Azure Cloud Architecture**
2. **Operations & Interfaces**
3. **IaC & GitOps**
4. **Wrap up & next steps**
5. **Q&A**

OD@Azure Architecture

Available Database Services



Oracle Exadata Database Service on Dedicated Infrastructure (19c, 26ai)



Oracle Exadata Database Service on Exascale Infrastructure (19c, 26ai)



Oracle Base Database Service (19c, 26ai)



Oracle Autonomous AI Database (19c, 26ai)

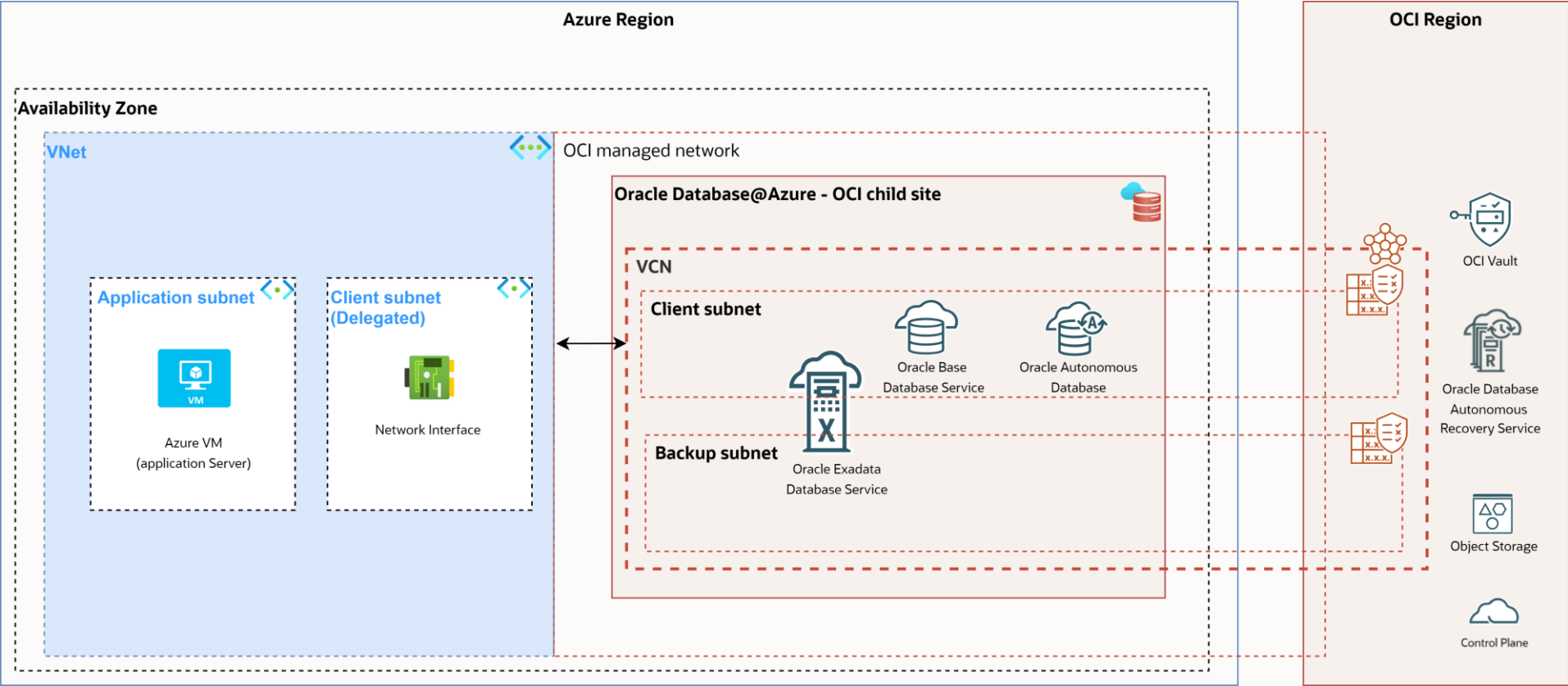


Oracle GoldenGate (21c, 26ai)

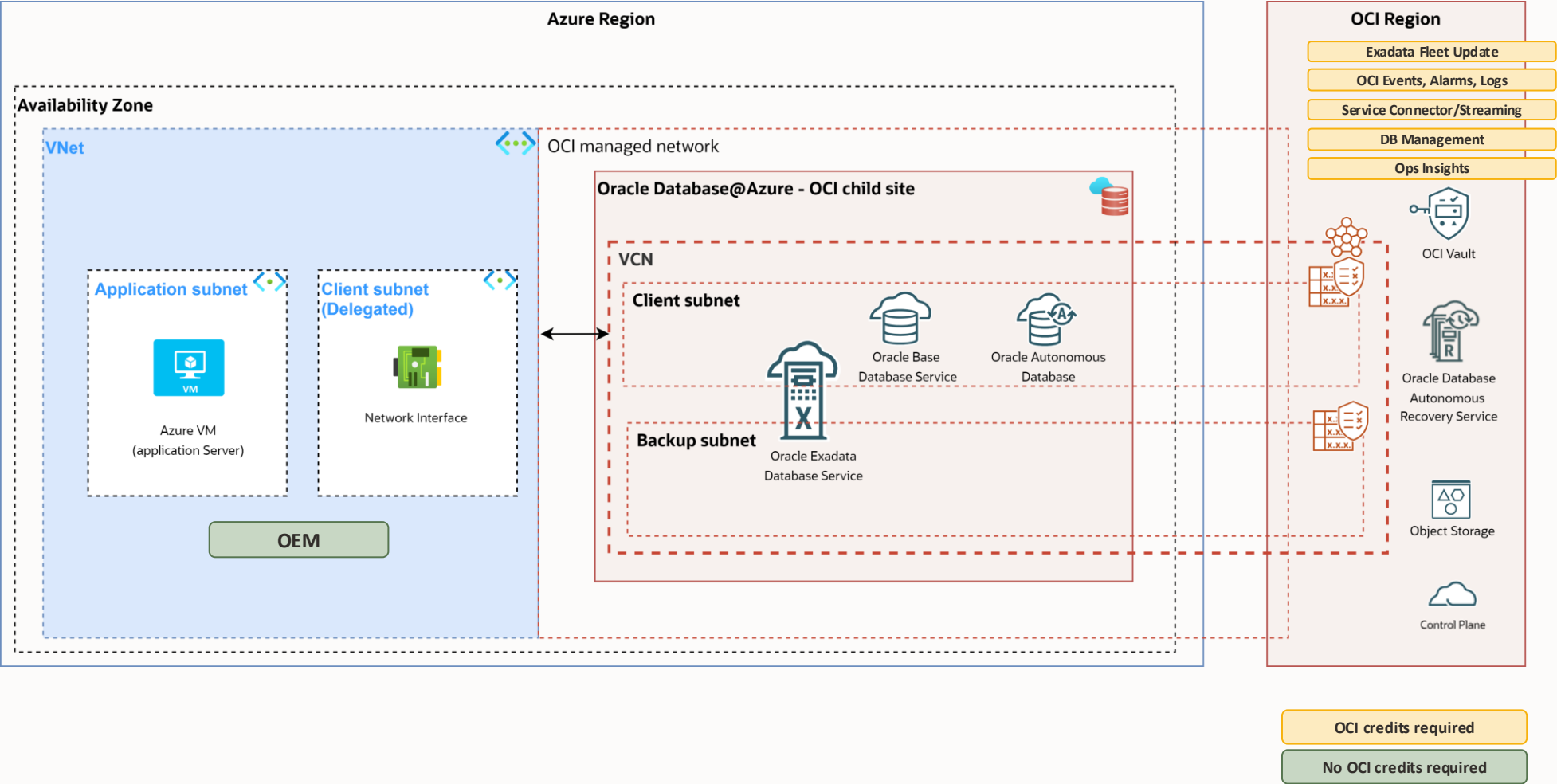


Oracle Database Autonomous Recovery Service

Architecture overview

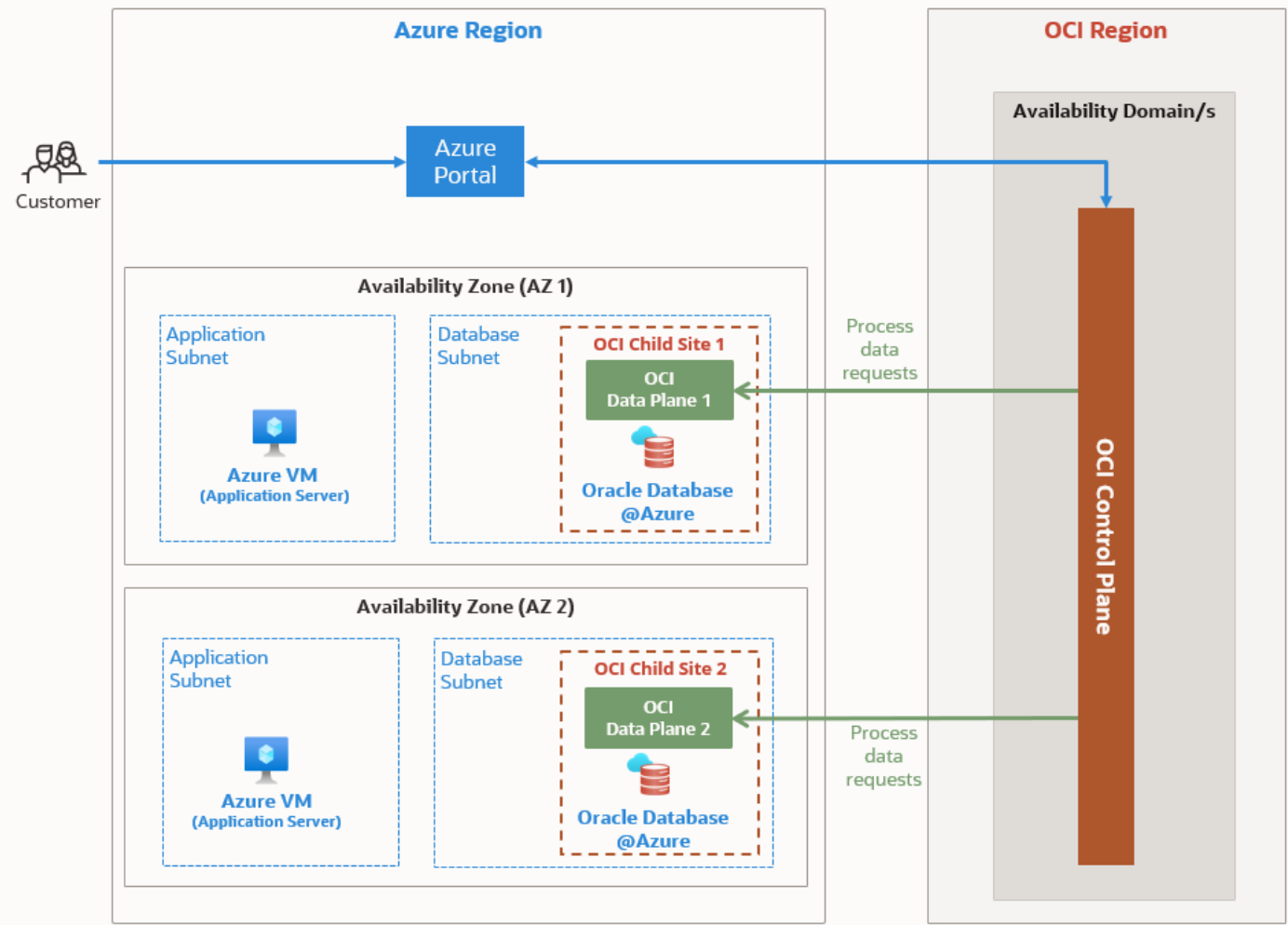


Optional operational services

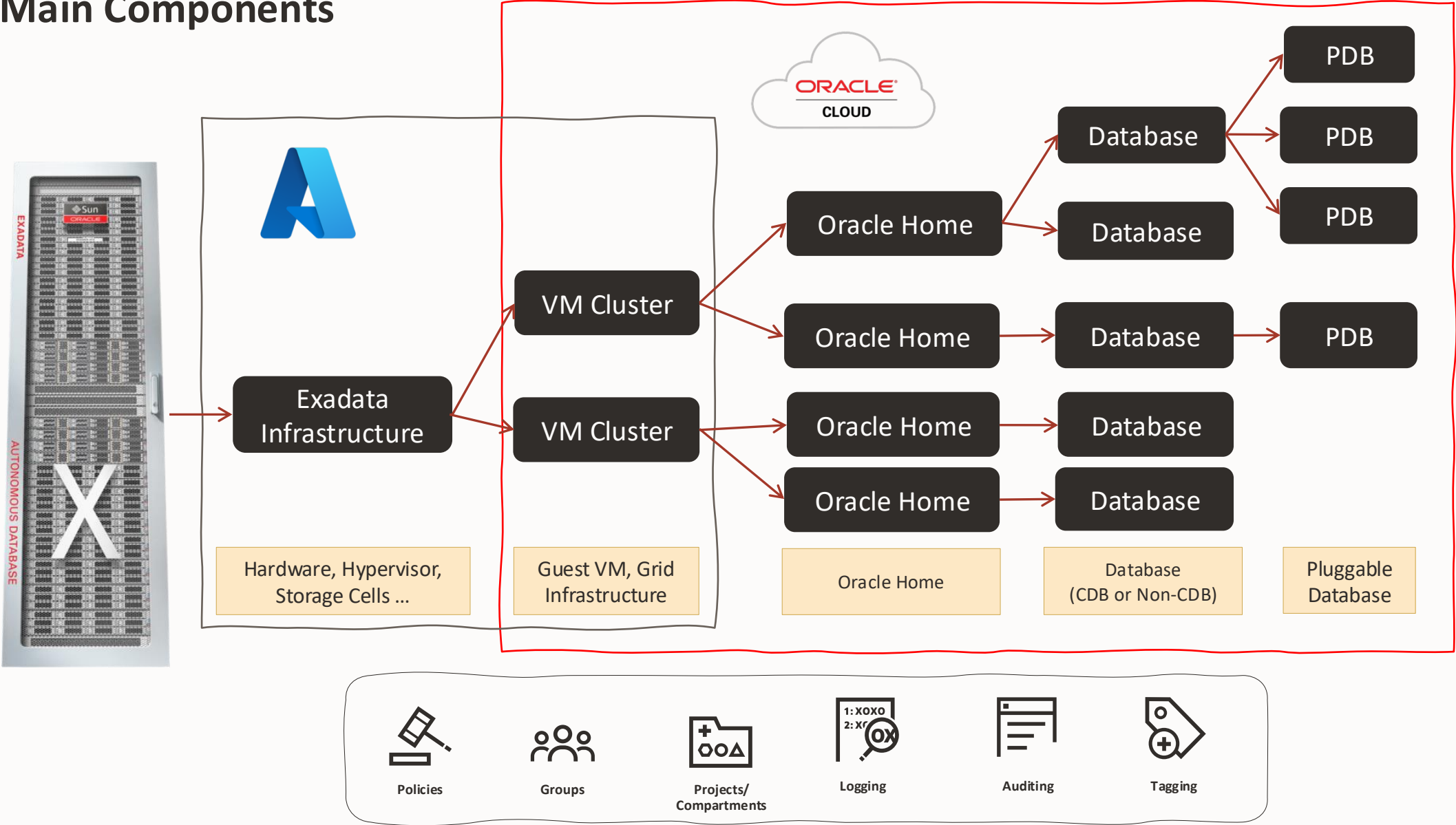


Operations & Interfaces

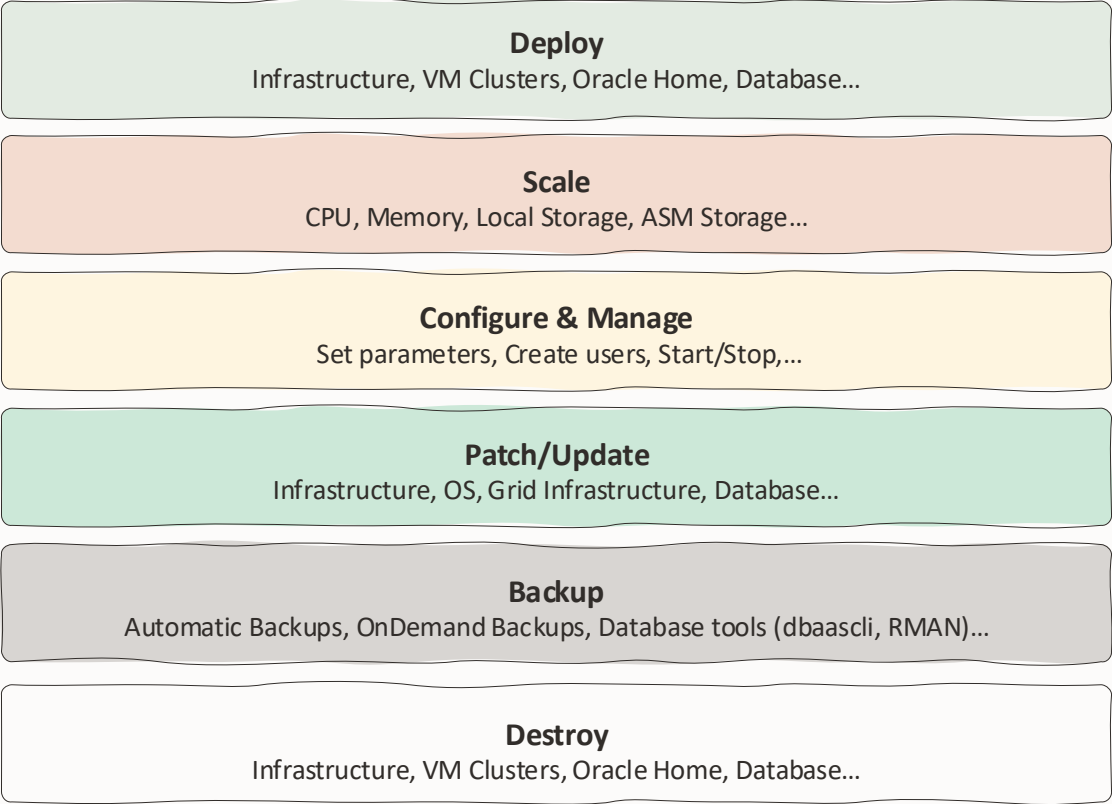
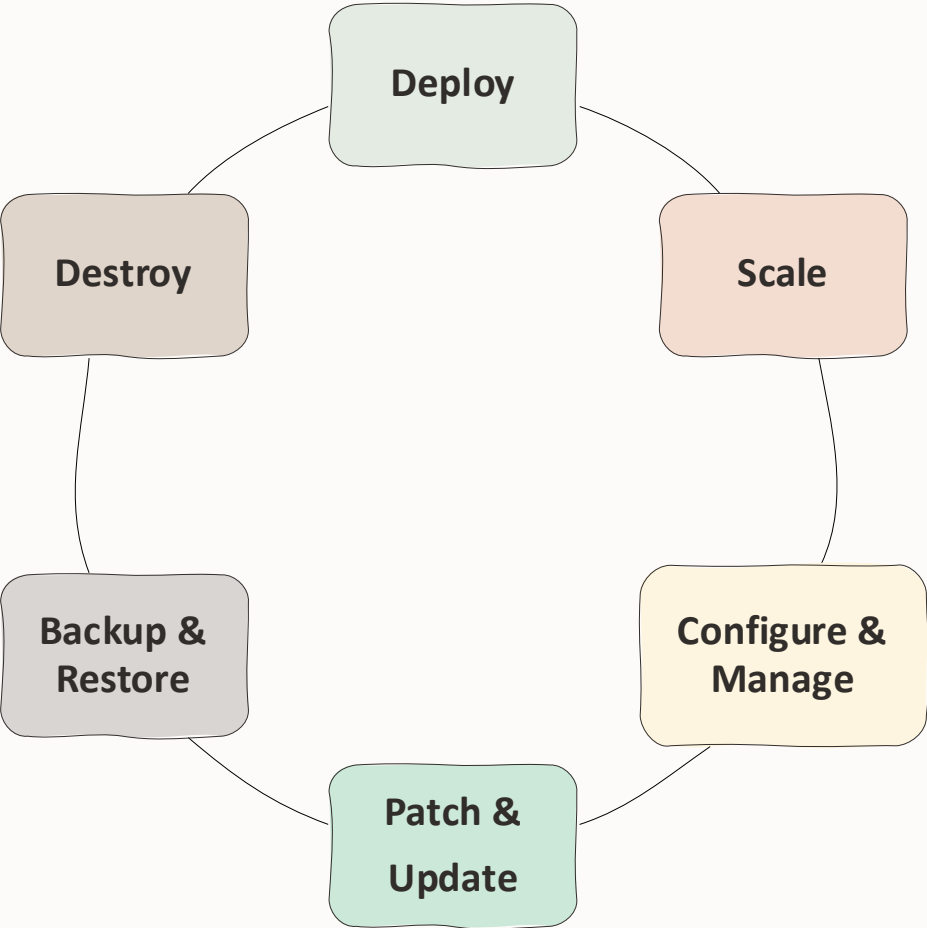
Control Plane Architecture



Main Components



Operations



Management Interfaces

Rest APIs

REST APIs that use HTTPS requests and responses for managing OCI and Azure resources, including Exadata Cloud

Consoles

Intuitive, graphical interfaces for creating and managing your OCI and Azure resources, are one of the interfaces to the Rest APIs

Command Line Interfaces (OCI CLI / Azure CLI)

Lightweight tools for command-line tasks. Support scripts and extend Console functionality

Software Development Kits (SDKs)

Enable custom solutions for Exadata Cloud and OCI and Azure services

DevOps Tools & Plug-ins

Tools like Terraform Provider and Ansible Collection for provisioning and automation

Cloud Shell

Browser-based terminal with pre-authenticated CLI, useful for tutorials and quick commands

Exadata - Local VM Command-Line Interfaces

CLI utilities located on the VM guests

Provisioned as part of the VM clusters on the Exadata Cloud Infrastructure, are available to perform various lifecycle and administration operations

dbaascli

Use the dbaascli utility to perform various database lifecycle and administration operations on the Exadata Cloud Infrastructure such as changing the password of a database user, starting a database, managing pluggable databases (PDBs), scaling the CPU core count in disconnected mode, and more.

ExaCli

Use the ExaCLI command-line utility to perform monitoring and management functions on Exadata storage servers in the Exadata Cloud.

bkup_api

Handles backup/recovery tasks, including on-demand database and PDB backups.

DEPRECATED

Must be connected to a VM in the Exadata Cloud VM cluster;
uses VM OS user security, not OCI user security.

Changes sync to OCI Control Plane via **DB Sync**.
Tools are auto-updated by Oracle on a regular basis.

Where I do what?



- Delegated Subnet
 - Creation/Deletion/Modify
 - View information
- Exadata infrastructure
 - Creation/deletion/Modify tags
 - View Information
- Exadata VM clusters
 - VM cluster creation/deletion/Modify tags
 - VMs status management/add/remove
 - View information
- *Oracle CDBs and PDBs aren't provisioned and managed on Azure but in OCI*



- Exadata infrastructure
 - Scale
 - Updates/Upgrades planning
- Exadata VM clusters
 - Configuration changes
 - Updates/Upgrades
 - GI management
 - Scale OCPU/Memory/Storage
- Oracle Home creation/deletion
- Oracle container databases and pluggable databases
 - Creation/Deletion
 - Status management
 - Updates/Upgrades
 - Backup & Recovery
 - DR setup and management
 - (...)

Management Interfaces vs. Operations

Operation	Azure				OCI					IaC	
	Console	az	Rest API	SDK	Console	OCI CLI	dbaascli	Rest API	SDK	Terraform ¹	Ansible Collections
Create/Delete Delegated Subnets	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗
Create/Delete Exadata Infrastructure instances	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗
Scale Exadata Infrastructure instances	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✗
Create/Delete Exadata/Exascale VM Cluster	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗
Stop/Start/Restart VMs	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✓
Create/Delete Database Oracle Homes	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Create/Delete Container Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Stop/Start/Restart Container Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Create/Delete/Clone Pluggable Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Stop/Start/Restart Pluggable Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Patch Guest OS/GI	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✓
Patch CDB/PDB	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Database Backup & Restore	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Scale VM OCPU/Memory/Storage (Local/Shared)	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✓
Schedule Infrastructure Maintenance	✗	✗	✗	✗	✓	✓	✗	✓	✓	✓	✗
Monthly Security Maintenance	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✗
Create software images	✗	✗	✗	✗	✓	✓	✗	✓	✓	✓	✓
Create/Delete Data Guard Association	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Perform Switchover/Failover	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓

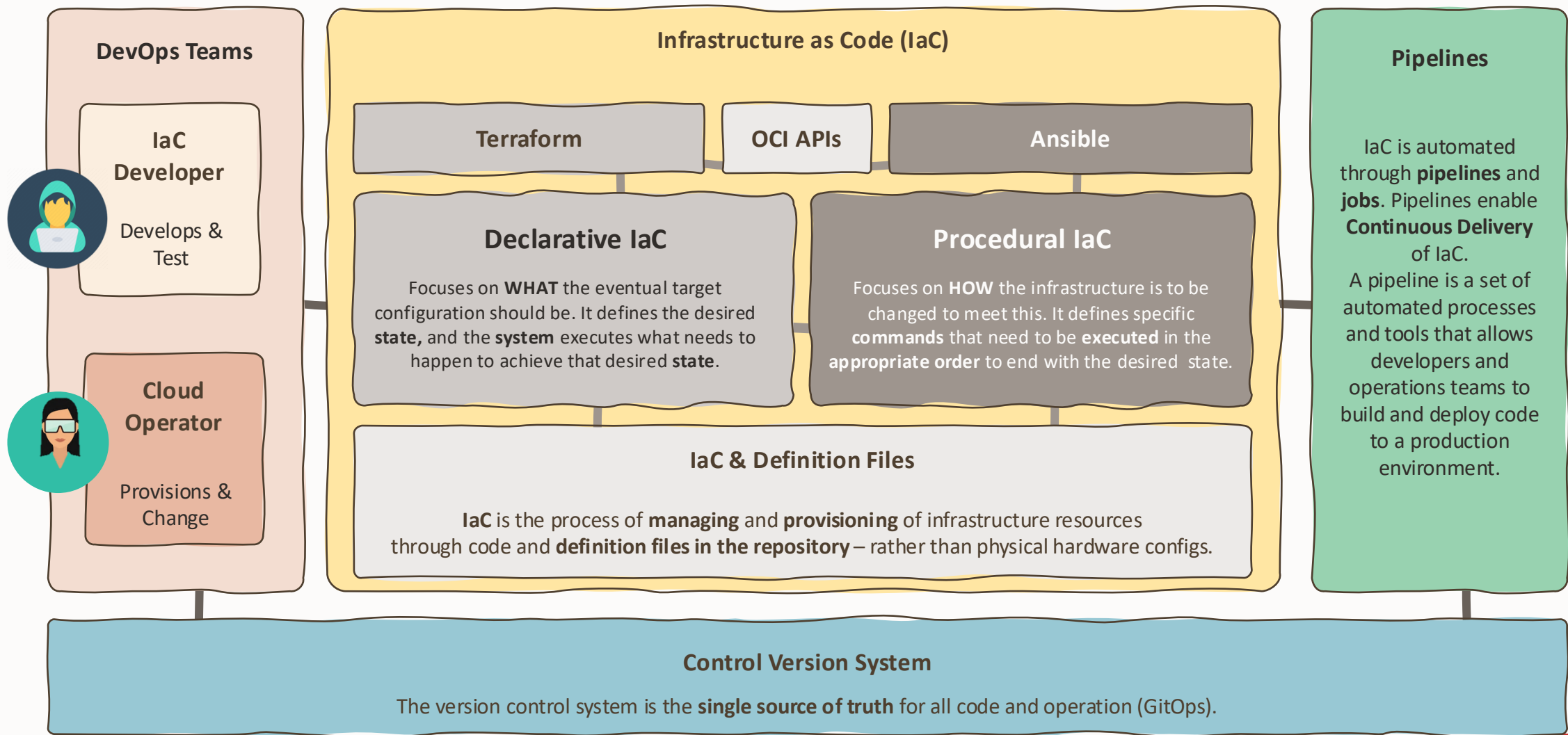


laC & GitOps

What is IaC?

*Infrastructure as Code is the process of **provisioning and managing** computing infrastructure through **code** instead of manual processes.*

IaC Technology Context & Personas



What is and does Terraform

Terraform is a **declarative** language that defines the **desired state** of infrastructure.

It uses a **state file** to consistently provision **Landing Zones (LZ)** and **workload** resources across **any cloud**.

Terraform is great to **provision the Landing Zone and workload** resources.



Declarative Language

Defines the "**what**", not the "how". A simple configuration expresses the entire desired state.



State Management

A **state file** acts as the **single source of truth**, mapping configurations to real-world resources.



Landing Zones Workload Resources

LZ set the **foundation** and **guardrails**.
Workload resources add the stacks on top across **any cloud**.

OCI Terraform Provider Resources



Network

[azurerm_virtual_network](#)
[azurerm_subnet](#)



Exadata Infrastructure

[azurerm_oracle_exadata_infrastructure](#)



VM Cluster

[azurerm_oracle_cloud_vm_cluster](#)



Autonomous DB

[azurerm_oracle_autonomous_database](#)



VM Cluster

[oci_database_cloud_vm_cluster](#)



DB Home

[oci_database_db_home](#)



CDB

[oci_database_database](#)



PDB

[oci_database_pluggable_database](#)



Terraform is not enough



The Limits

Terraform has some boundaries. It cannot manage tasks inside a resource, only the resource itself.

-  **No patching:** Not for OS/DB/app patching or upgrades
-  **No in-guest config:** Doesn't install packages, edit files, or manage services on existing hosts
-  **No app/data config:** doesn't manage DB schemas/users or app settings.
-  **Non-Updatable** changes (needs to re-deploy)
-  **Last resort only:** Provisioners (remote-exec/local-exec) and generic helpers (terraform_data, deprecated null_resource) aren't suited for Config Management and can leak secrets to state/logs

The Solution

Use the [lifecycle statement](#) block to **bridge the gap** between **declarative IaC** and **procedural** realities.

```
lifecycle {
  ignore_changes = [
    database[0].admin_password,      <--- Ignoring by default
    database[0].database_software_image_id <--- ignore db_home patching
    database_software_image_id,      <--- ignore db_home patching
    db_version,                      <--- ignore db_home patching
  ]
}
```

The [ignore_changes](#) skip diffs for specific fields (config edits or drift)

What is Ansible

Ansible is a declarative, **YAML-based** automation engine, **agentless** IaC tool used primarily for **configuration management**.



Agent less

- **No** permanent **agents** on managed systems.
- Uses **temporary** network **connections** via **SSH** or Windows Remote Management



Vs. Alternatives

- Chef, Puppet and Salt require installed agents.
- **Ansible eliminates** this **complexity** and overhead .



Minimum Requirement

- **Only requires Python** installed on managed systems.
- Nothing more.



Central Orchestration

- **Centralized** and efficient **architecture**.
- All **orchestration** is done from a **control node** (masterd host).



SSH Security

- **Leverages** native **SSH security** without additional ports or running daemons

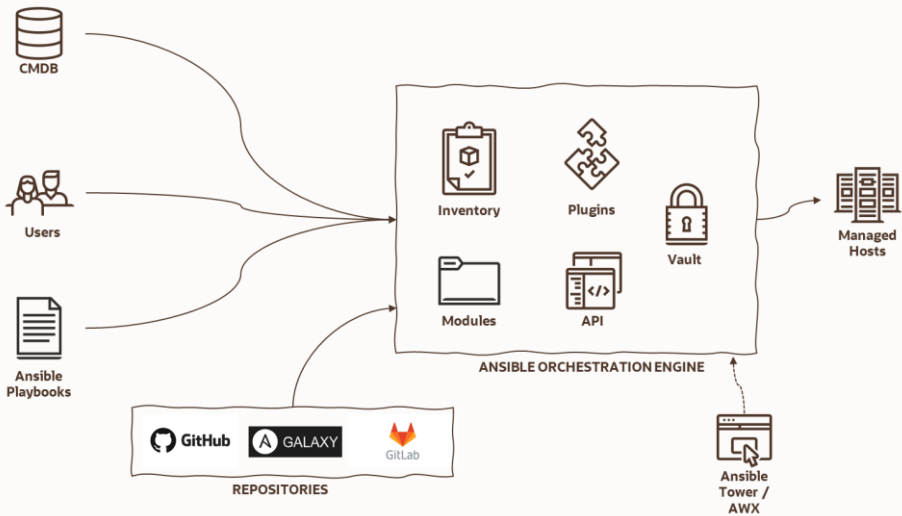
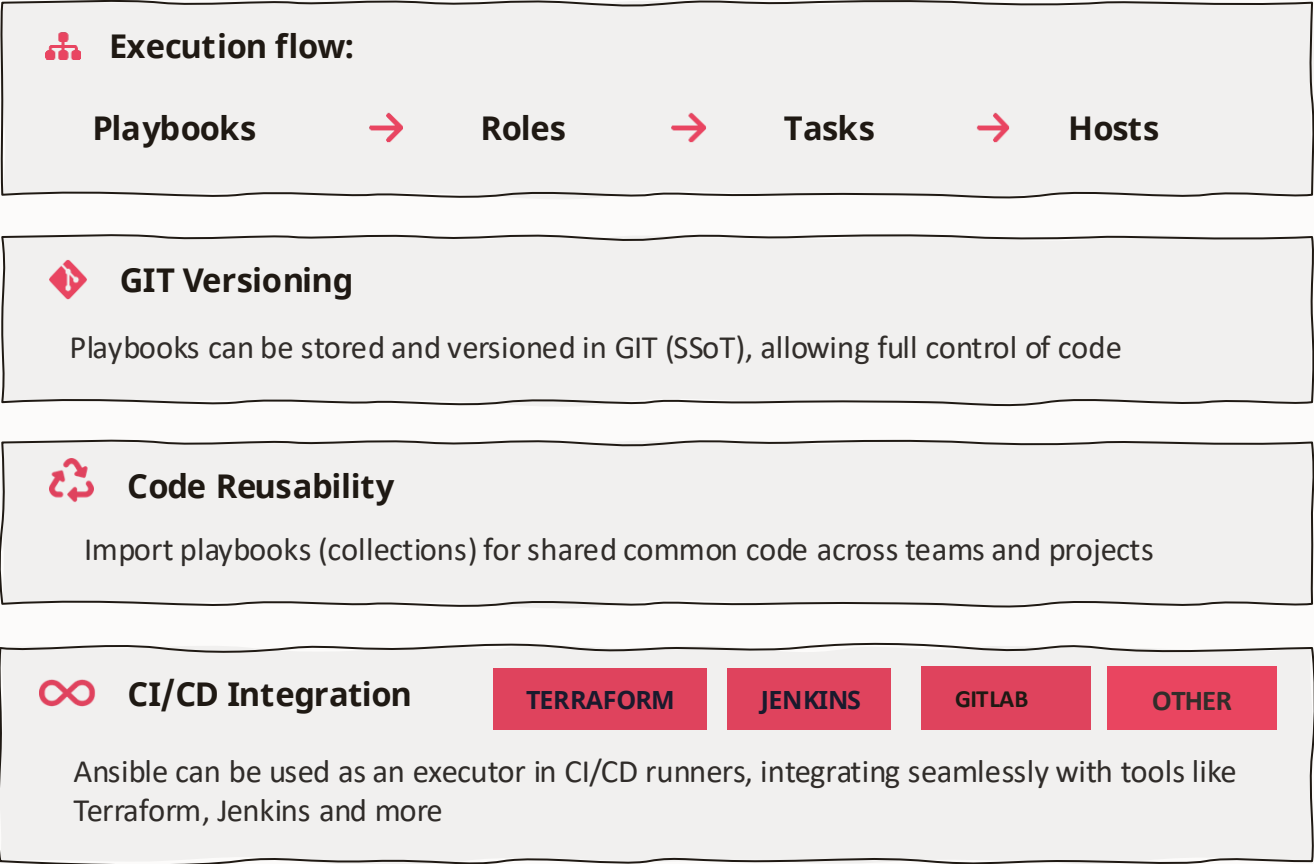


Zero Overhead

- **Zero resource consumption** on managed servers when no tasks are running

Playbooks: The Heart of Ansible

Playbooks are the language by which Ansible orchestrates, configures, administers, or deploys systems.



OCI Ansible collection

Ansible Collection helps you **automate the provisioning and configuring** of Oracle Cloud Infrastructure services and resources



Github repository

[OCI Ansible collection](#)



Ansible Tower/ AWX Integration

[Ansible Tower](#)

[AWX.](#)



Ansible Inventory Plugin

[Static Inventory](#) & [Dynamic inventory](#)



Supported Services

- The Oracle Cloud Infrastructure Ansible Collection provides **hundreds of [modules](#)** to **automate** provisioning and configuration of **OCI resources**.



Minimal Requirements

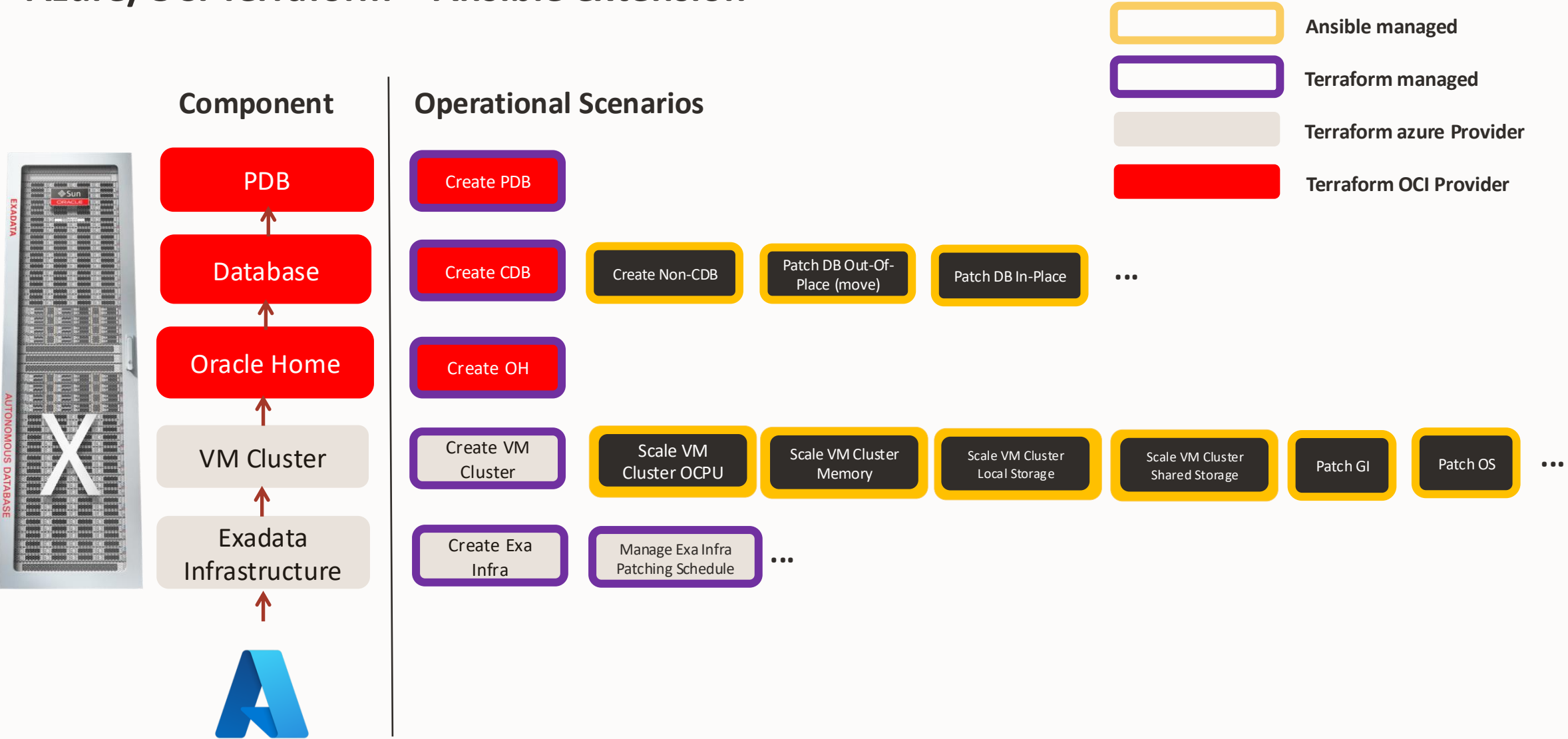
- The OCI **Ansible collection** requires Ansible version. **2.9 or later**



OCI Collection key benefits

- Provision & configure OCI via Ansible.
- Dynamic inventory for discovery.
- Centralized runs (AWX/Tower, OLAM).
- **Broad, Oracle-maintained coverage.**

Azure/OCI Terraform + Ansible extension

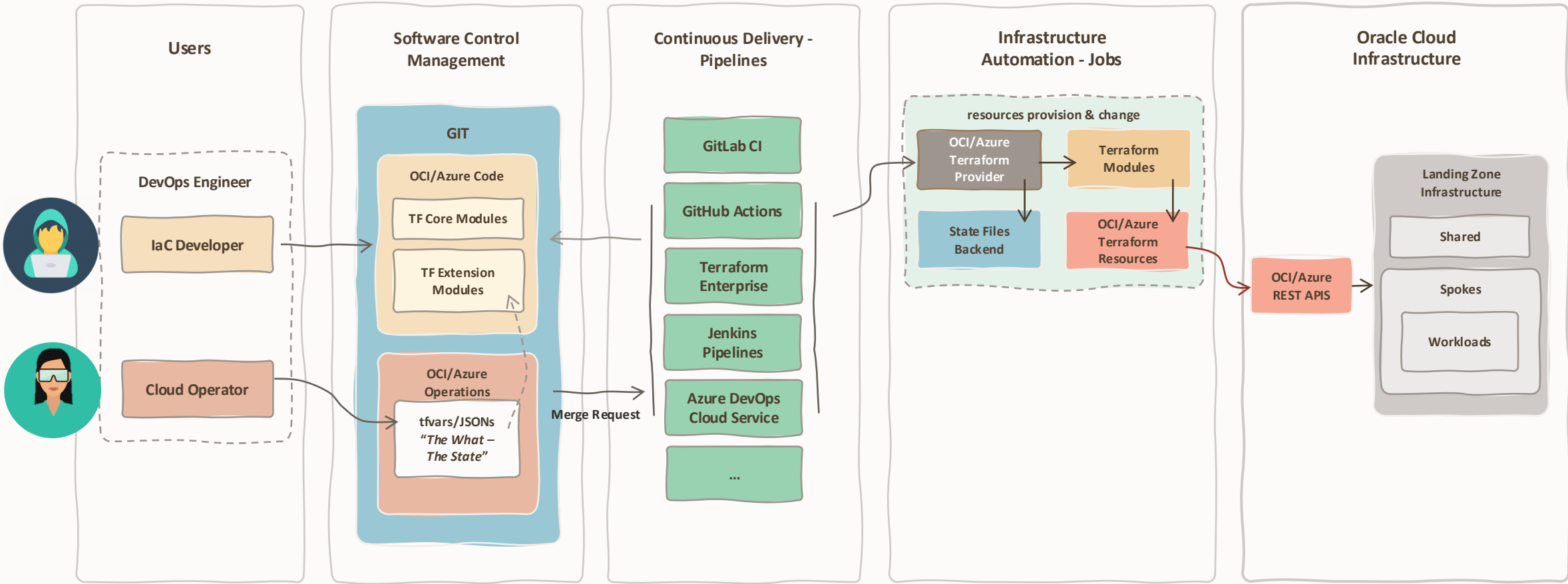
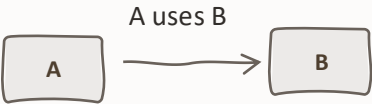


What is GitOps?

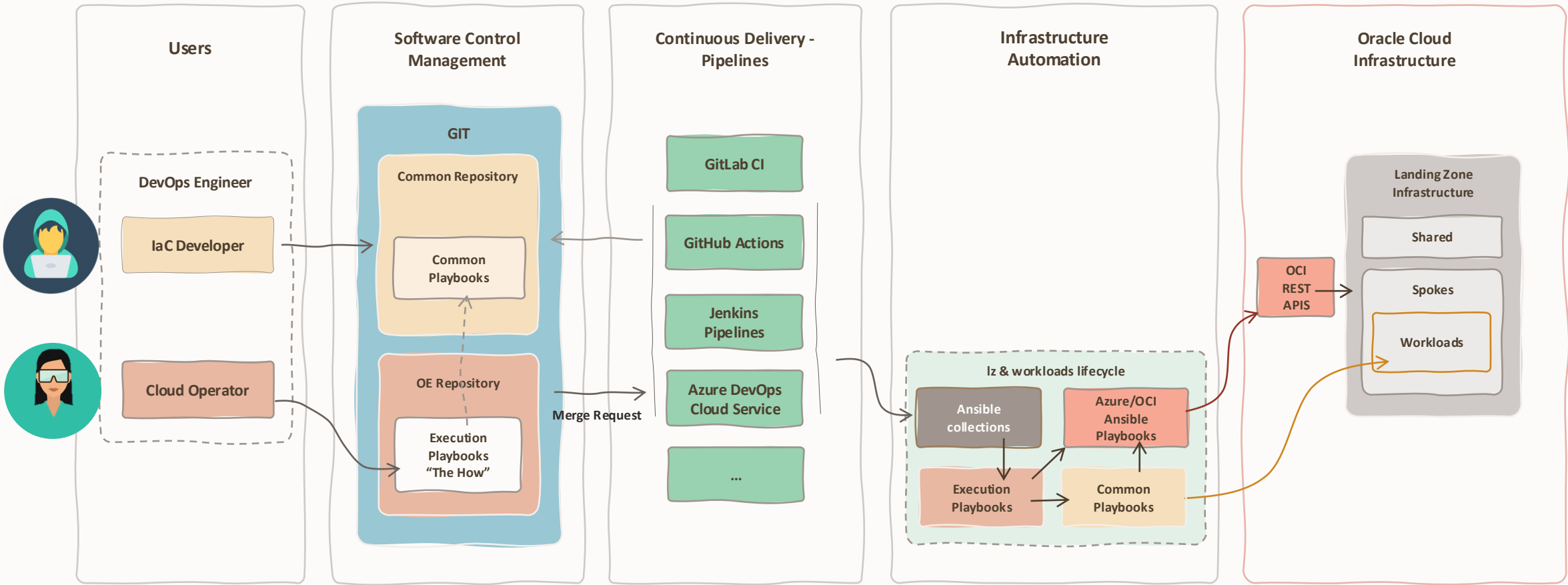
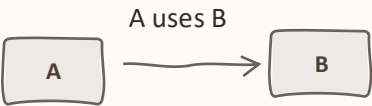
*GitOps is an operational practice that **applies DevOps best practices**, like **version control, collaboration, compliance**, and **CI/CD** to infrastructure automation.*

GitOps becomes a great common multi-cloud operating model.

GitOps using a Declarative Pattern with Terraform



GitOps using a Procedural Pattern with Ansible



Wrap up & Next Steps

Takeaways | Next steps

- There is ***not a single tool to manage everything.***
- Different management interfaces to manage ExaDB-D components, ***leverage to your preferred automation tooling.***
- The use of a ***GitOps*** approach ***simplifies your operations.***
- ***Exadata Fleet Update facilitates*** the schedule, patching process and automation needed.
- OCI Basic monitoring & integration services can be used to forward valuable information to your IT Systems (SIEM, ITSM, etc.).

Additional Operations Workshops

➤ Exadata Fleet Update

Learn what is Exadata Fleet Update and how it can be used to patch Exadata OS, GI & DBs at scale

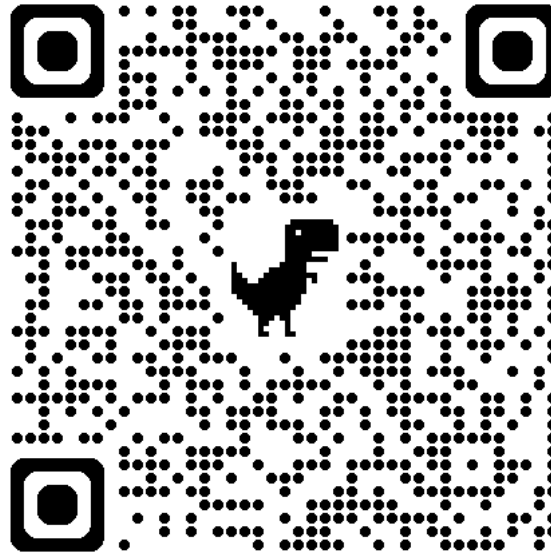
➤ OD@ExaDB-D Operations Deep Dive

Learn how you can use different management interfaces to provision, change, maintain & manage high availability for Exadata Cloud Service in azure

➤ GitOps Multi Cloud Operating Model Deep Dive

GitOps uses Git as the single source of truth to automate infrastructure and application delivery. This Infrastructure-as-Code (IaC) model streamlines operations, enhances governance, and scales across any cloud.

Stay tuned



[Operations Advisory](#)
[GitHub Repository](#)

Thank you



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